

1. Electric potential

We have seen in earlier lectures that the electric field at a position $\vec{r}$ due to any configuration of charges can be represented as the gradient of a scalar valued position function. This position function is labeled as the potential at position $\vec{r}$ due to the electric field.

$$\vec{E}(\vec{r}) = -\vec{\nabla}_{\vec{r}}\Phi(\vec{r})$$

Here, the function $\Phi(\vec{r})$ at position $\vec{r}$ is the electric potential due to the electric field $\vec{E}$.

We present a few more scenarios where we show that the electric field, generated due to different configuration of charges, can at any position be written as a gradient of a scalar function. For the first case we take an arbitrary configuration of charges and in the second case we choose a single charge Q_A .

$$\begin{aligned}\vec{E}_{\rho}(\vec{r}) &= \iiint_V \frac{1}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \rho(\vec{r}') d^3\vec{r}' \\ &= \iiint_V \frac{1}{4\pi\epsilon_0} \left\{ -\vec{\nabla}_{\vec{r}} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \right\} \rho(\vec{r}') d^3\vec{r}' \\ &= -\vec{\nabla}_{\vec{r}} \left\{ \iiint_V \frac{1}{4\pi\epsilon_0} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \rho(\vec{r}') d^3\vec{r}' \right\}\end{aligned}$$

$$\Phi(\vec{r}) = \iiint_V \frac{1}{4\pi\epsilon_0} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \rho(\vec{r}') d^3\vec{r}'$$

$$\begin{aligned}\vec{E}_{Q_A}(\vec{r}_B) &= \frac{1}{4\pi\epsilon_0} Q_A \frac{\vec{r}_B - \vec{r}_A}{|\vec{r}_B - \vec{r}_A|^3} \\ &= \frac{1}{4\pi\epsilon_0} Q_A \left\{ -\vec{\nabla}_{\vec{r}_B} \left(\frac{1}{|\vec{r}_B - \vec{r}_A|} \right) \right\} \\ &= -\vec{\nabla}_{\vec{r}_B} \left\{ \frac{1}{4\pi\epsilon_0} Q_A \left(\frac{1}{|\vec{r}_B - \vec{r}_A|} \right) \right\}\end{aligned}$$

$$\Phi(\vec{r}_B) = \frac{1}{4\pi\epsilon_0} Q_A \left(\frac{1}{|\vec{r}_B - \vec{r}_A|} \right)$$

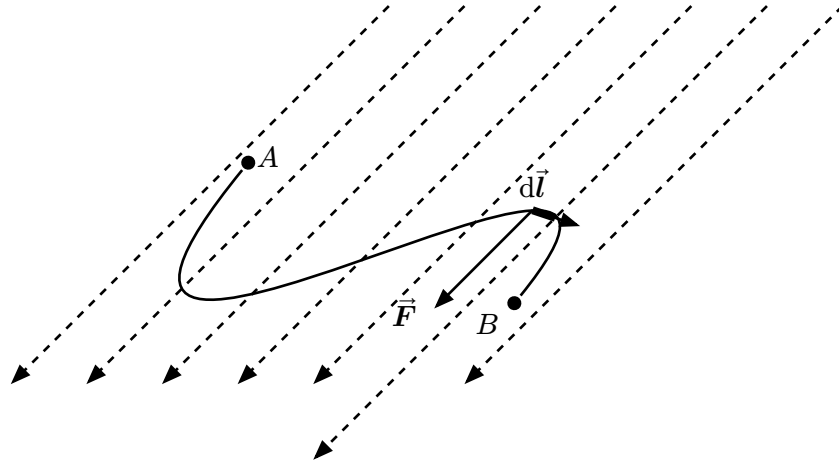


Figure 1: The figure shows Electric Field lines as dashed lines with an arrow head indicating their direction. The path taken by a test charge from point A to point B. The figure also shows the force acting on the test charge and the infinitesimal distance the charge is moving.

The force acting on a point charge q at position $\vec{r}$ due to the electric field $\vec{E}$ can be written as

$$\vec{F}(\vec{r}) = q\vec{E}(\vec{r})$$

Therefore the work done by an external force to move a point charge q from point A to point B can be written as

$$W = - \int_{\vec{r}_A}^{\vec{r}_B} \vec{F}(\vec{r}) \cdot d\vec{l}(\vec{r})$$

The negative sign appears because the work is done by an external agent against the force acting on the charge due to the electric field.

Now that we have all the tools required to define the potential. We formally define the electric potential, denoted by Φ , at position $\vec{r}$ as the work done by an external agent to move a unit positive test charge from infinity to the position $\vec{r}$ without accelerating the charge. The potential at position $\vec{r}$ is then given by

$$\Phi(\vec{r}) = - \int_{\infty}^{\vec{r}} \vec{E}(\vec{r}') \cdot d\vec{r}'$$

The potential is measured in Volts (V) where, 1 Volt = 1 Joule per Coulomb.

Let us now consider the scenario in Figure 1 where a test charge q is being moved from point A to point B. Let us compute the work done by an external agent against the electrostatic force to move the charged particle from point A to B.

$$\begin{aligned} W &= - \int_{\vec{r}_A}^{\vec{r}_B} \vec{F}(\vec{r}) \cdot d\vec{l}(\vec{r}) \\ &= -q \int_{\vec{r}_A}^{\vec{r}_B} \vec{E}(\vec{r}) \cdot d\vec{l}(\vec{r}) \end{aligned}$$

Now we know the electric field can be written as the negative gradient of the potential and so we have

$$\begin{aligned}
W &= - \int_{\vec{r}_A}^{\vec{r}_B} \vec{F}(\vec{r}) \cdot d\vec{l}(\vec{r}) \\
&= -q \int_{\vec{r}_A}^{\vec{r}_B} \vec{E}(\vec{r}) \cdot d\vec{l}(\vec{r}) \\
&= q \int_{\vec{r}_A}^{\vec{r}_B} \vec{\nabla}_{\vec{r}} \Phi(\vec{r}) \cdot d\vec{l}(\vec{r}) \\
&= q \int_{\vec{r}_A}^{\vec{r}_B} d\Phi(\vec{r}) \\
&= q\{\Phi(\vec{r}_B) - \Phi(\vec{r}_A)\}
\end{aligned}$$

We now have the work done by the external agent against the electric field to be

$$W = q\{\Phi(\vec{r}_B) - \Phi(\vec{r}_A)\}$$

We can see that the work depends only on the initial and final position of the test charge. So we can see that the work done against the electric field which in this case, as the charge is moving with no acceleration, is equal to the work done by the electric field in absolute magnitude is independent of the path connective point A and B through which the charge was moved.

We now look at the work done when moving the charge from position B back to position A in a completely different path as

$$- \int_{\vec{r}_B}^{\vec{r}_A} q\vec{E}(\vec{r}) \cdot d\vec{l}(\vec{r}) = q\{\Phi(\vec{r}_A) - \Phi(\vec{r}_B)\}$$

Therefore we can write for the round trip in different paths from point A to point B and then back to point A as

$$\oint \vec{E}(\vec{r}) \cdot d\vec{l}(\vec{r}) = 0$$

So for the electrostatic force the work done by the force on any test charge is independent of the path taken by the charge, is zero for a closed loop path, and can be represented by the negative gradient of a scalar function. We define any such force as a *Conservative Force* and the corresponding field as a *Corresponding Field*.